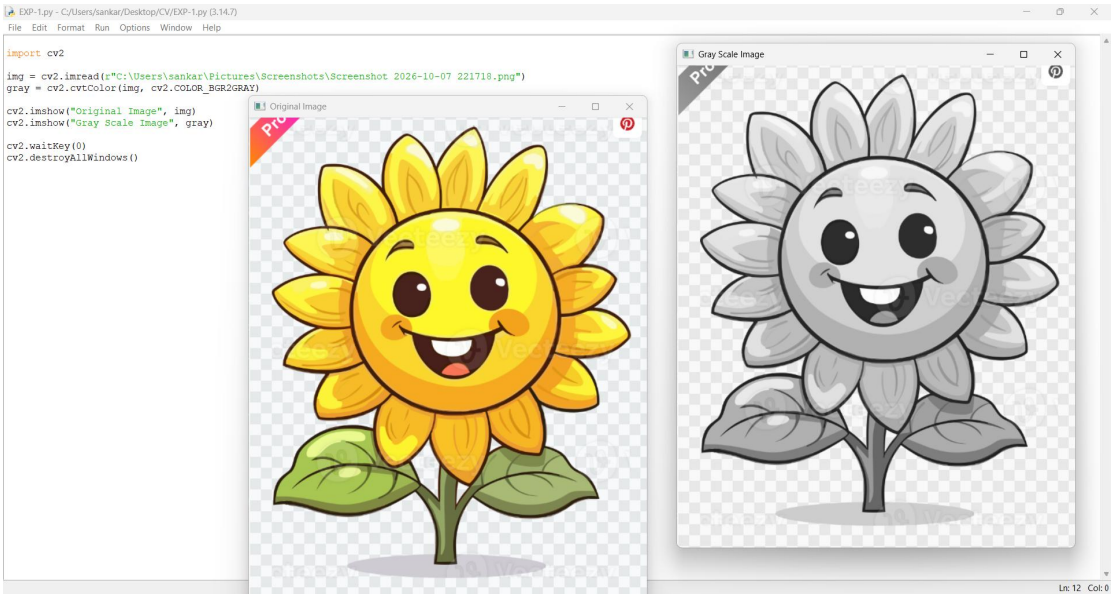
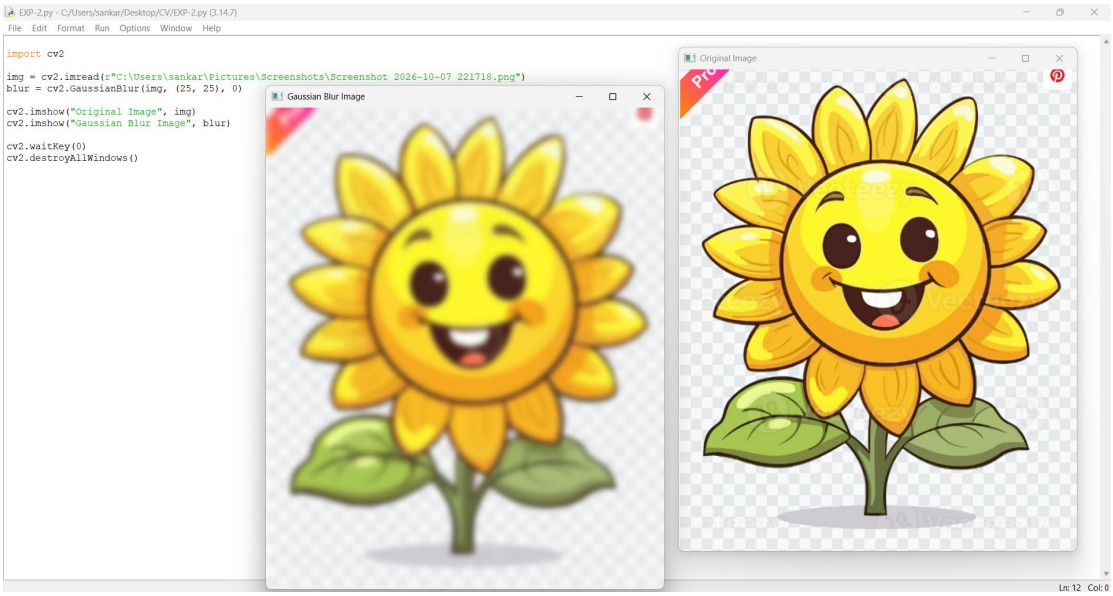
LAB OUTPUTS
ITA0513
Computer Vision

Name : K.Niharika
Register No: 192411124

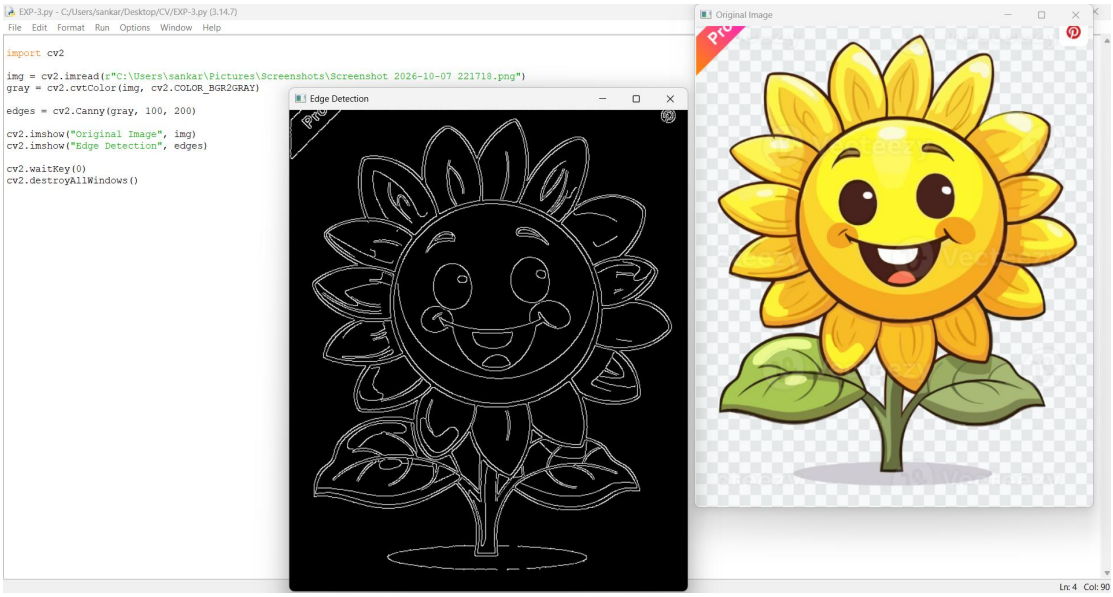
EXP1-Image Reading and Conversion into Gray Scale using OpenCV



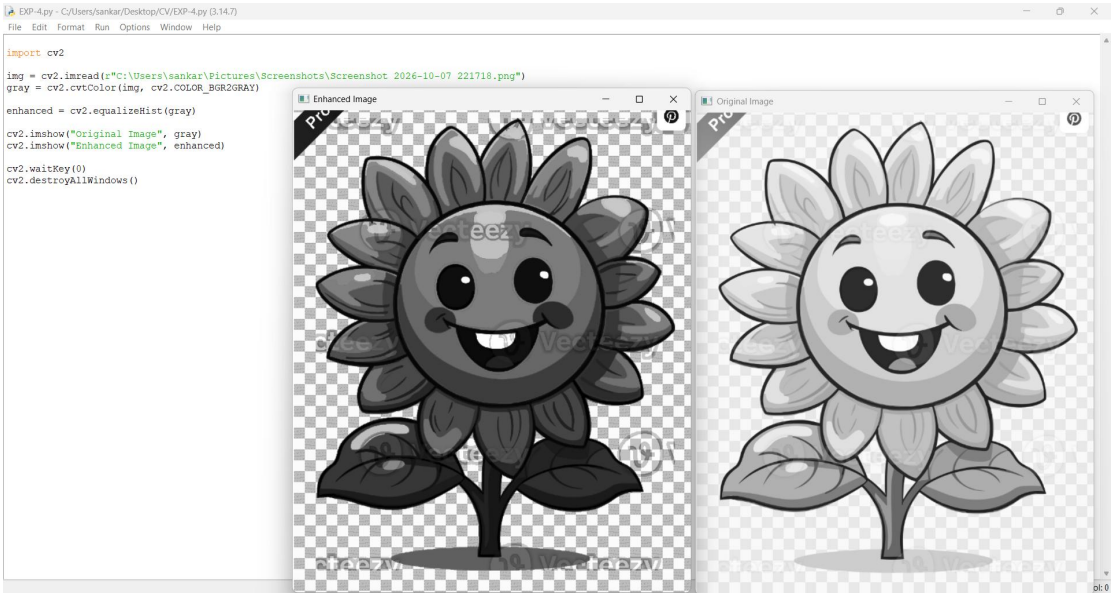
EXP2-Image Blurring using Gaussian Blur Operation



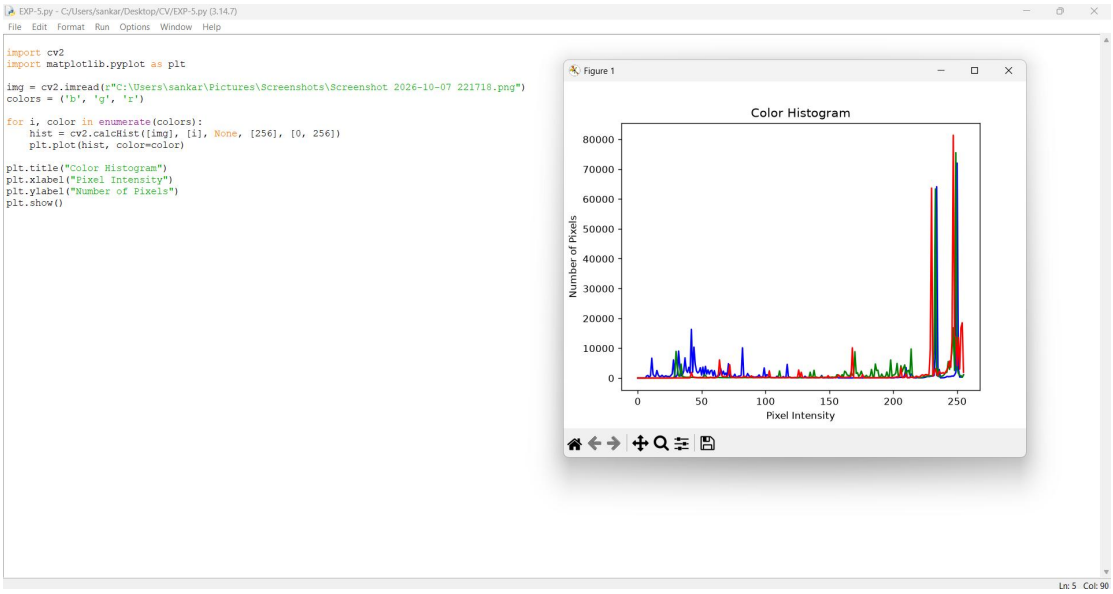
EXP-3:Image Outline Detection using Canny Edge Detection



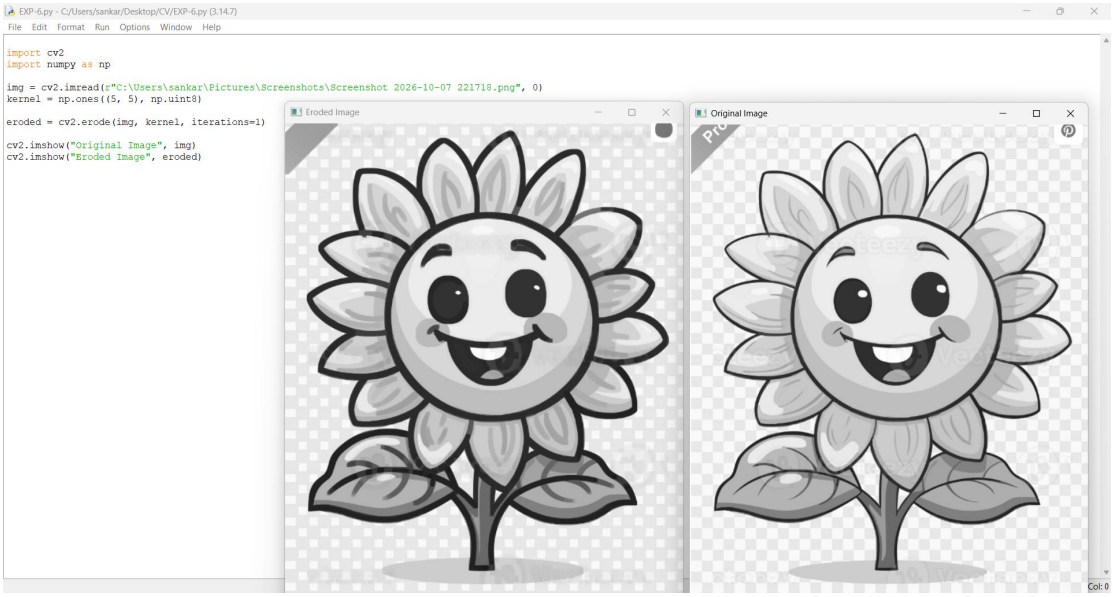
EXP-4:Histogram Equalization and Image Enhancement using OpenCV



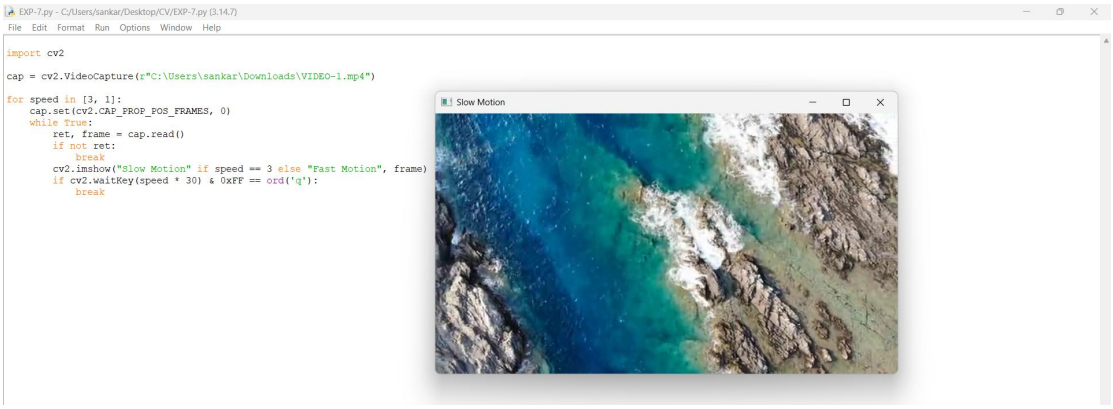
EXP-5:Histogram Analysis of Image Based on Color Levels using OpenCV



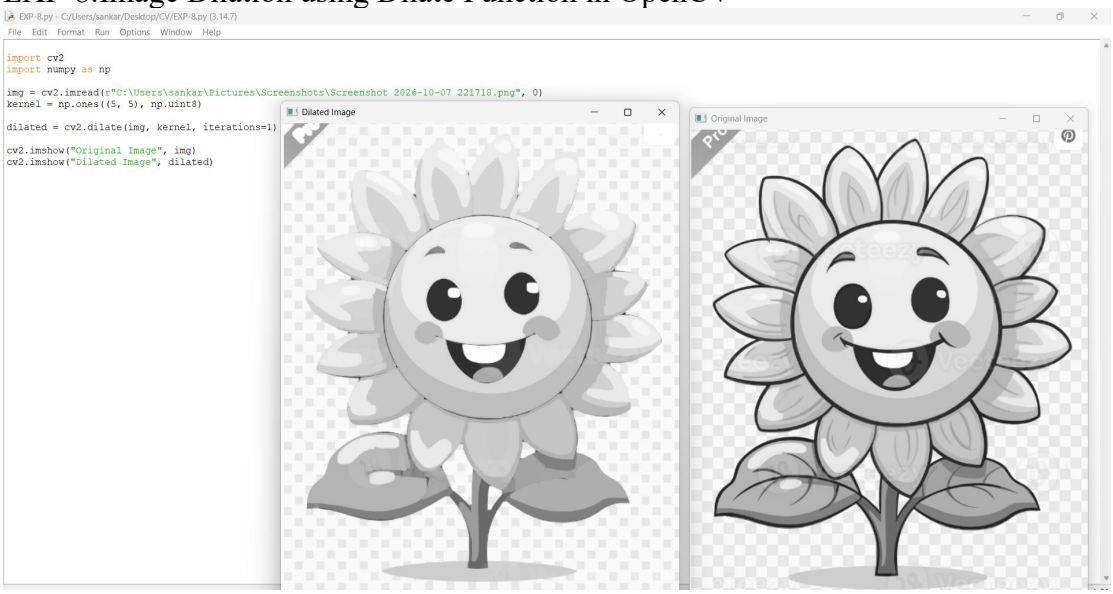
EXP-6:Image Erosion using Erode Function in OpenCV



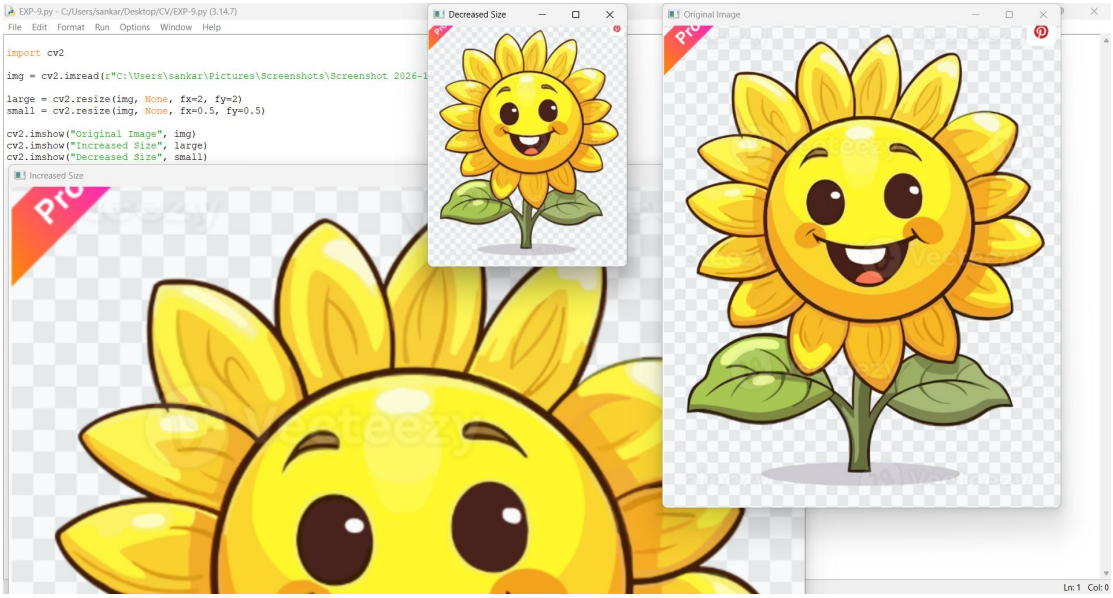
EXP-7:Video Processing using OpenCV- Slow Motion and Fast Motion



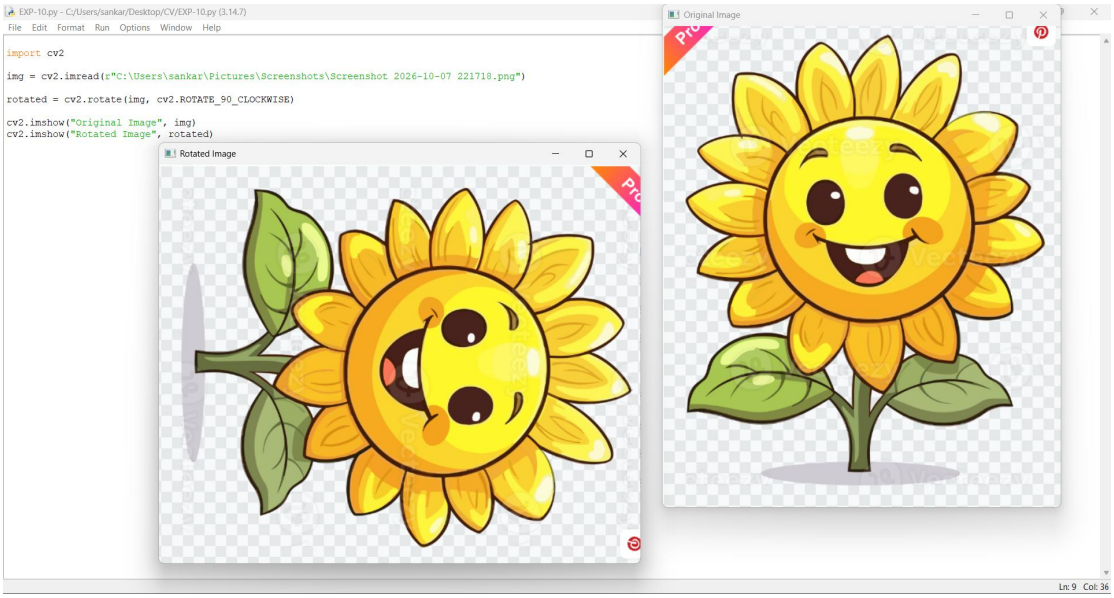
EXP-8:Image Dilation using Dilate Function in OpenCV



EXP-9:Image Scaling Techniques for Increasing and Decreasing Image Size using OpenCV



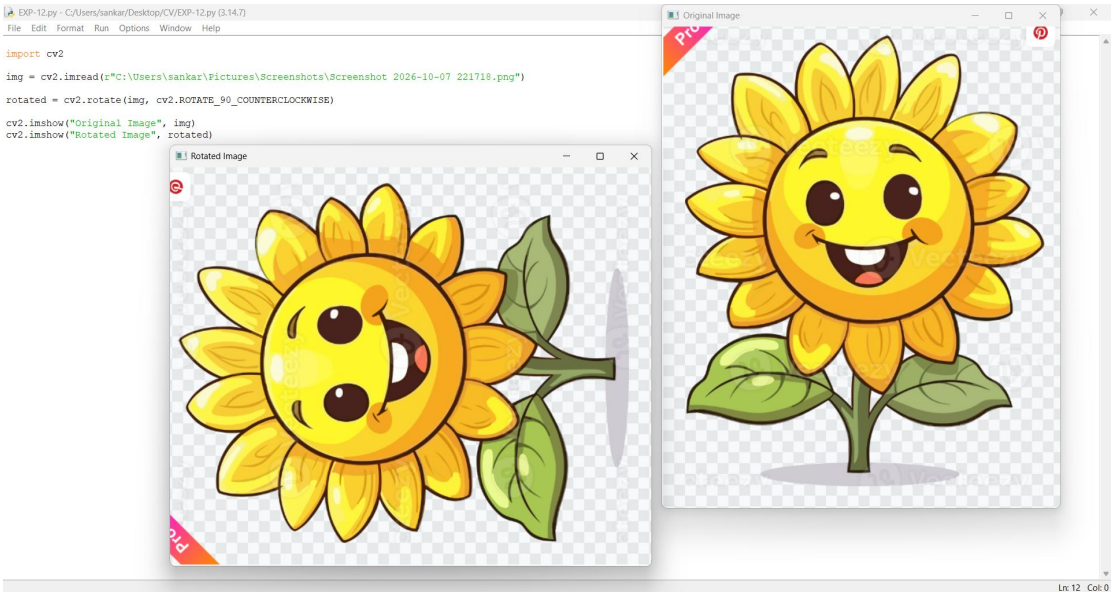
EXP-10: 90-Degree Clockwise Rotation of Image using OpenCV



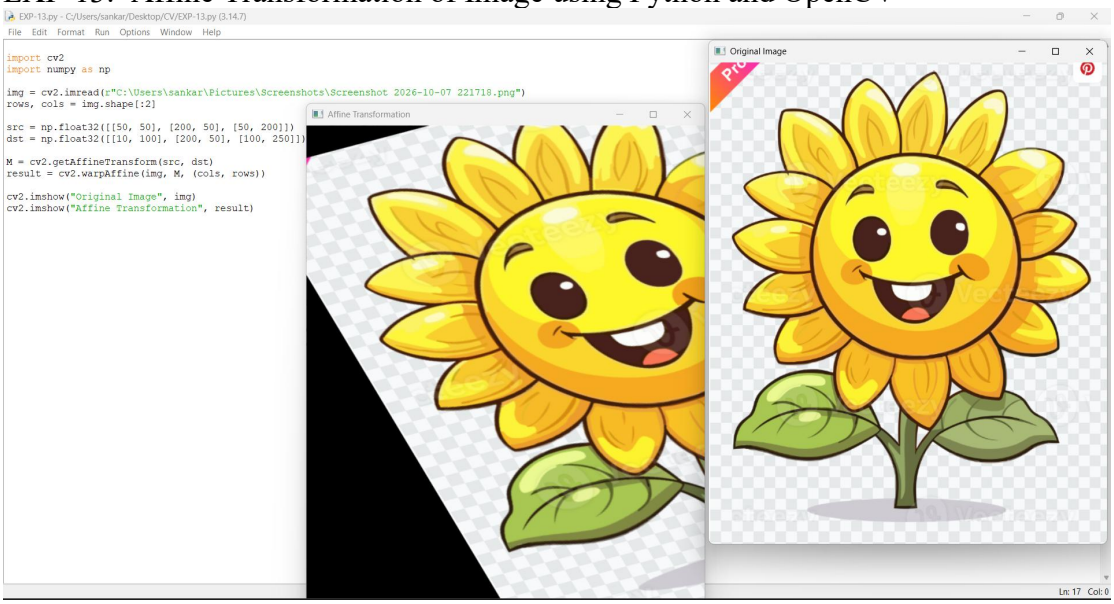
EXP-11: 180-Degree Clockwise Rotation of Image using OpenCV



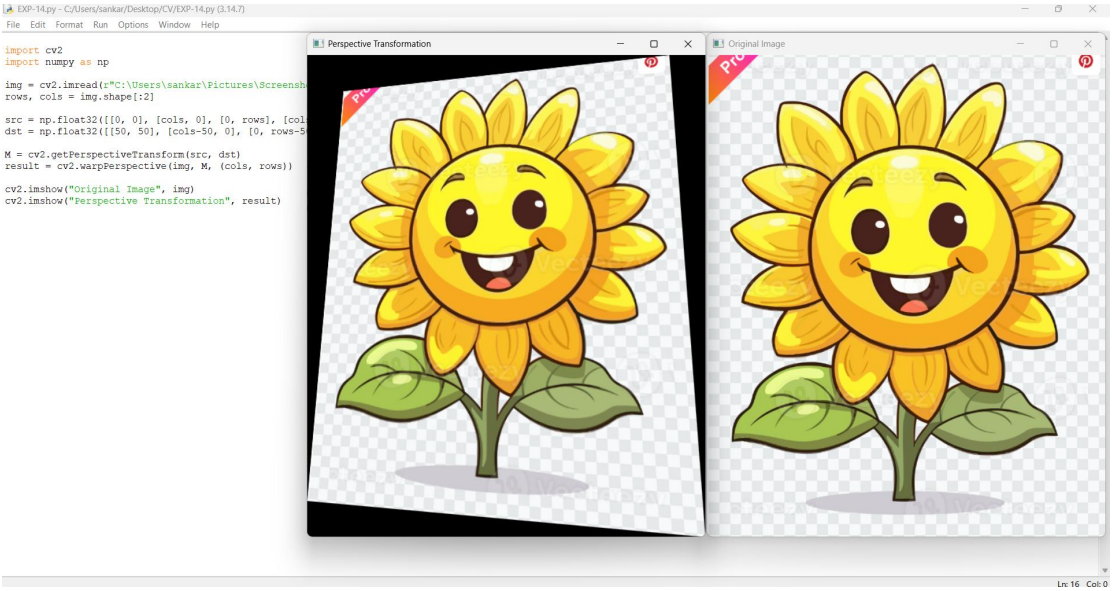
EXP-12: 270-Degree Clockwise Rotation of Image using OpenCV



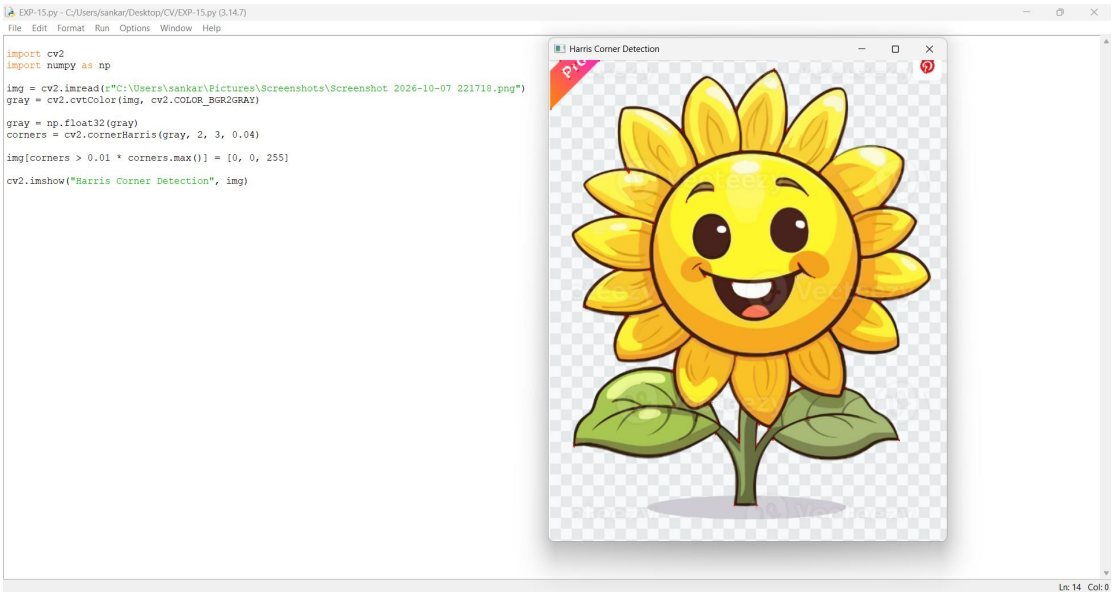
EXP-13: Affine Transformation of Image using Python and OpenCV



EXP-14: Perspective Transformation of Image using Python and OpenCV



EXP-15: Corner Detection using Harris Corner Detection Algorithm



EXP 16: Implementation of Sobel Edge Detection Algorithm using OpenCV

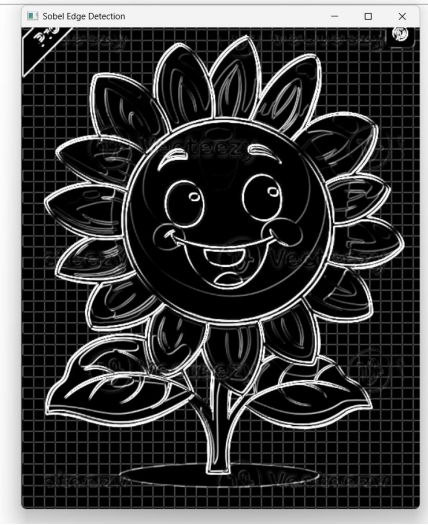
```
import cv2

img = cv2.imread("C:/Users/sankar/Pictures/Screenshots/Screenshot 2026-10-07 221718.png", 0)

x = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3)
y = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3)

edge = cv2.convertScaleAbs(x + y)

cv2.imshow("Sobel Edge Detection", edge)
```



EXP-17: Implementation of Image Watermarking Technique using OpenCV

```
import cv2

img = cv2.imread("C:/Users/sankar/Pictures/Screenshots/Screenshot 2026-10-07 221718.png")
cv2.putText(img, "WATERMARK", (50, 50),
            cv2.FONT_HERSHEY_SIMPLEX, 1,
            (0, 0, 255), 2)

cv2.imshow("Watermarked Image", img)
```



EXP-18: Image Cropping, Copying and Pasting using ROI in OpenCV

```
import cv2

img = cv2.imread("C:/Users/sankar/Pictures/Screenshots/Screenshot 2026-10-07 221718.png")

roi = img[50:150, 50:150]
img[200:300, 200:300] = roi.copy()

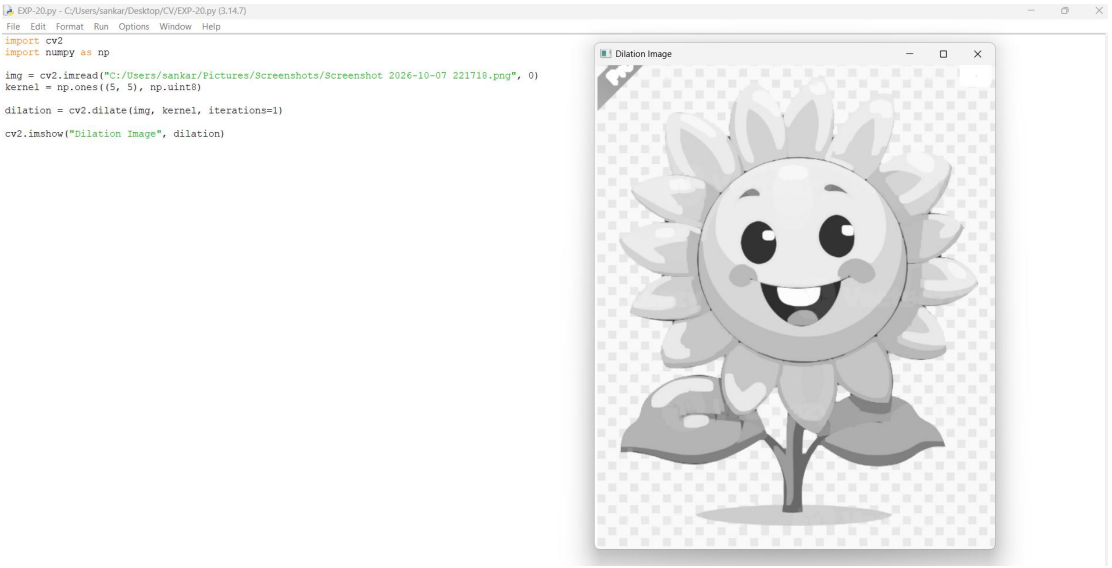
cv2.imshow("ROI Image", img)
```



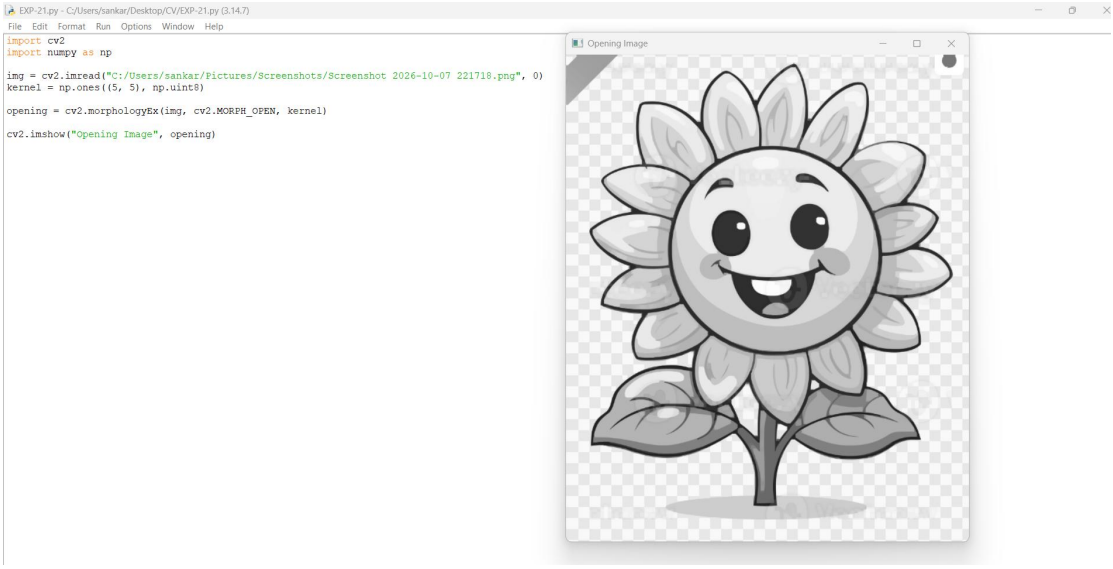
EXP-19: Implementation of Erosion Morphological Operation using OpenCV



EXP-20: Implementation of Dilation Morphological Operation using OpenCV



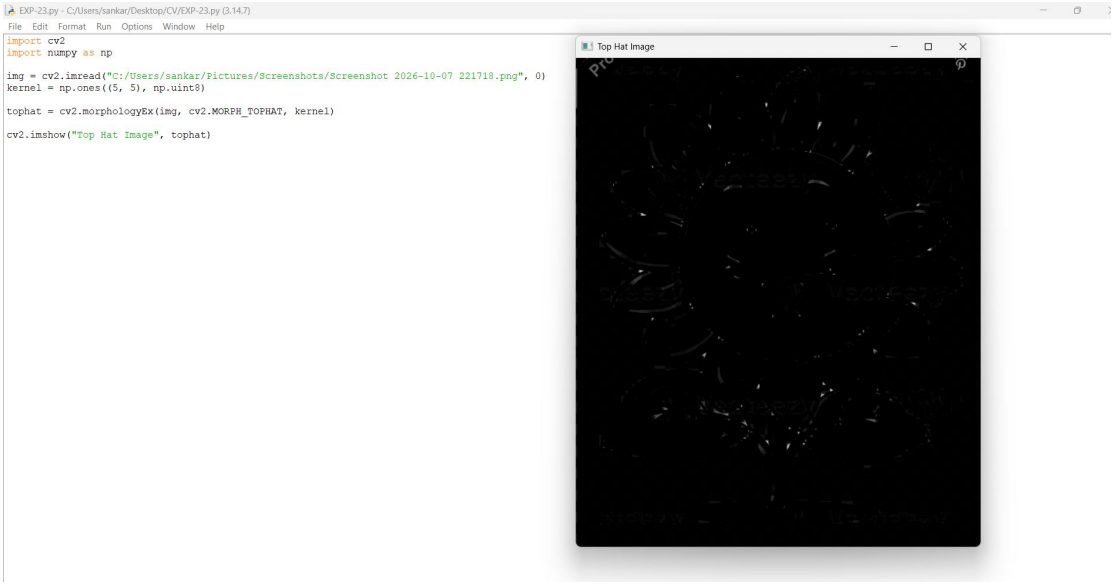
EXP-21 : Implementation of Morphological Opening Operation using OpenCV



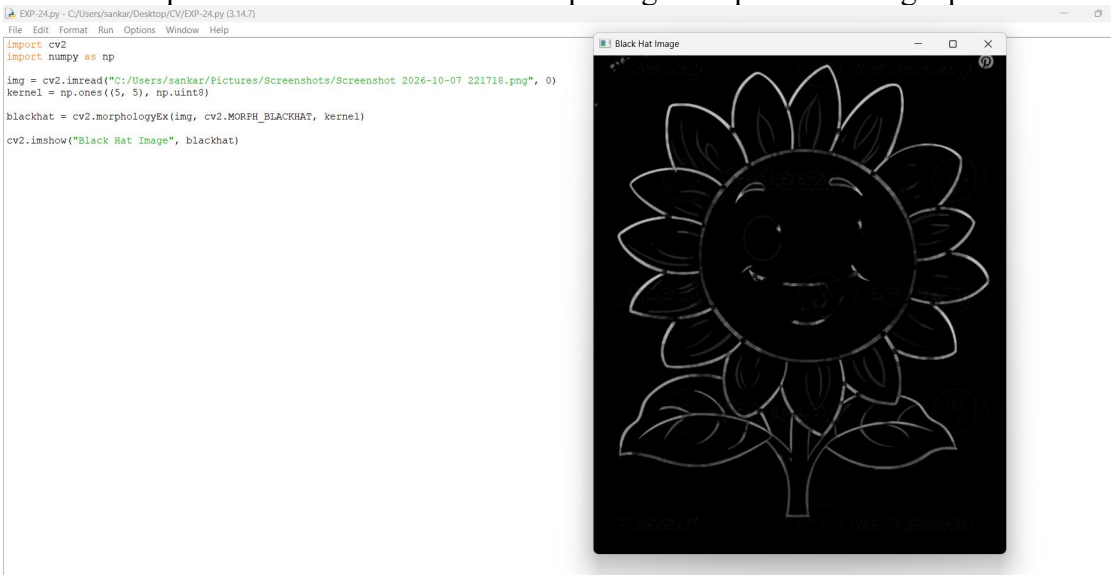
EXP-22 : Implementation of Morphological Closing Operation using OpenCV



EXP-23: Implementation of Top Hat Morphological Operation using OpenCV



EXP-24 : Implementation of Black Hat Morphological Operation using OpenCV



EXP-25 : Object Recognition from Image using OpenCV

```
EXP-25.py - C:/Users/sankar/Desktop/CV/EXP-25.py (3.14.7)
File  Edit  Format  Run  Options  Window  Help
import cv2

img = cv2.imread("C:/Users/sankar/Pictures/Screenshots/Screenshot 2026-10-07 221718.png")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

_, thresh = cv2.threshold(gray, 127, 255, cv2.THRESH_BINARY)
contours, _ = cv2.findContours(thresh, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)

cv2.drawContours(img, contours, -1, (0, 255, 0), 2)

cv2.imshow("Object Recognition", img)
```



EXP-26: Implementation of Reverse Video Frames using OpenCV

```
EXP-26.py - C:/Users/sankar/Desktop/CV/EXP-26.py (3.14.7)
File  Edit  Format  Run  Options  Window  Help
import cv2

cap = cv2.VideoCapture(r"C:\Users\sankar\Downloads\VIDEO-1.mp4")

if not cap.isOpened():
    print("Error: Video not found!")
    exit()

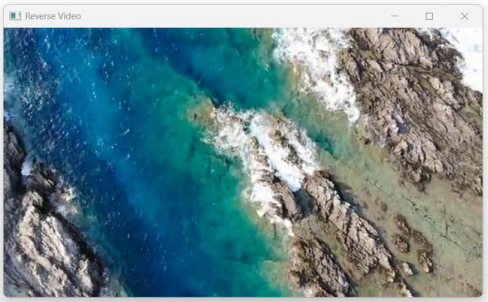
frames = []

while True:
    ret, frame = cap.read()
    if not ret:
        break
    frames.append(frame)

cap.release()
print("Total frames:", len(frames))

for frame in frames[::-1]:
    cv2.imshow("Reverse Video", frame)
    if cv2.waitKey(30) & 0xFF == ord('q'):
        break

cv2.destroyAllWindows()
```



EXP-27: Implementation of Human Face Detection using OpenCV

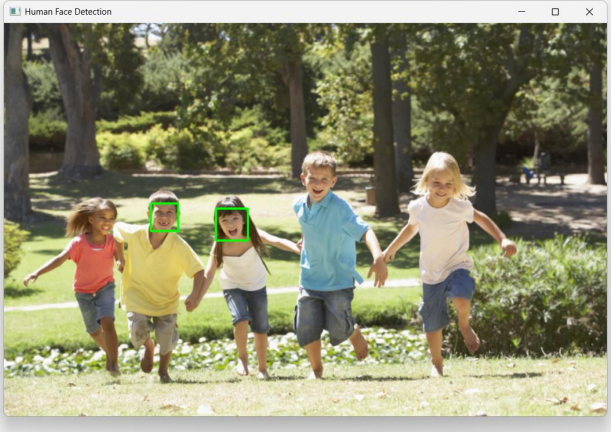
```
EXP-27.py - C:/Users/sankar/Desktop/CV/EXP-27.py (3.14.7)
File  Edit  Format  Run  Options  Window  Help
import cv2

img = cv2.imread(r"C:\Users\sankar\Pictures\Screenshots\Screenshot 2026-10-22 103512.png")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

face = cv2.CascadeClassifier(cv2.data.haarcascades + "haarcascade_frontalface_default.xml")
faces = face.detectMultiScale(gray, 1.1, 5)

for (x, y, w, h) in faces:
    cv2.rectangle(img, (x, y), (x+w, y+h), (0, 255, 0), 2)

cv2.imshow("Human Face Detection", img)
```



EXP-28: Implementation of Vehicle Detection in Video using OpenCV

```
import cv2
from ultralytics import YOLO

model = YOLO("yolov8n.pt")
cap = cv2.VideoCapture(r"C:\Users\sankar\Downloads\VIDEO-1.mp4")

while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break

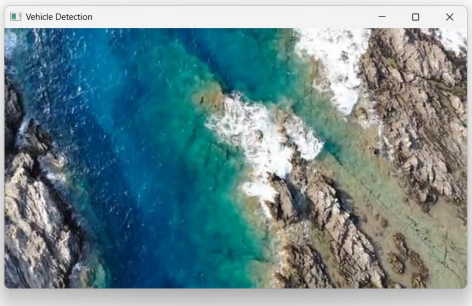
    results = model(frame, verbose=False)

    for result in results:
        for box in result.boxes:
            cls = int(box.cls[0])
            name = model.names[cls]

            if name in ["car", "bus", "truck", "motorcycle"]:
                x1, y1, x2, y2 = map(int, box.xyxy[0])
                cv2.rectangle(frame, (x1, y1), (x2, y2), (0, 255, 0), 2)
                cv2.putText(frame, name, (x1, y1-10),
                            cv2.FONT_HERSHEY_SIMPLEX, 0.6, (0, 255, 0), 2)

    cv2.imshow("Vehicle Detection", frame)

    if cv2.waitKey(1) & 0xFF == ord('q'):
        break
```



EXP-29: Implementation of Human Eye Detection using OpenCV

```
import cv2

img = cv2.imread(r"C:\Users\sankar\Pictures\Screenshots\Screenshot 2026-10-28 125924.png")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

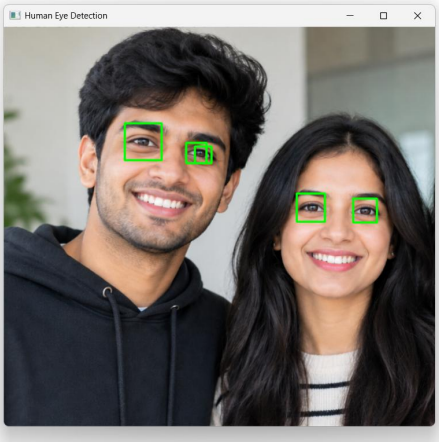
eye = cv2.CascadeClassifier(cv2.data.haarcascades + "haarcascade_eye.xml")
face = cv2.CascadeClassifier(cv2.data.haarcascades + "haarcascade_frontalface_default.xml")

faces = face.detectMultiScale(gray, 1.1, 5)

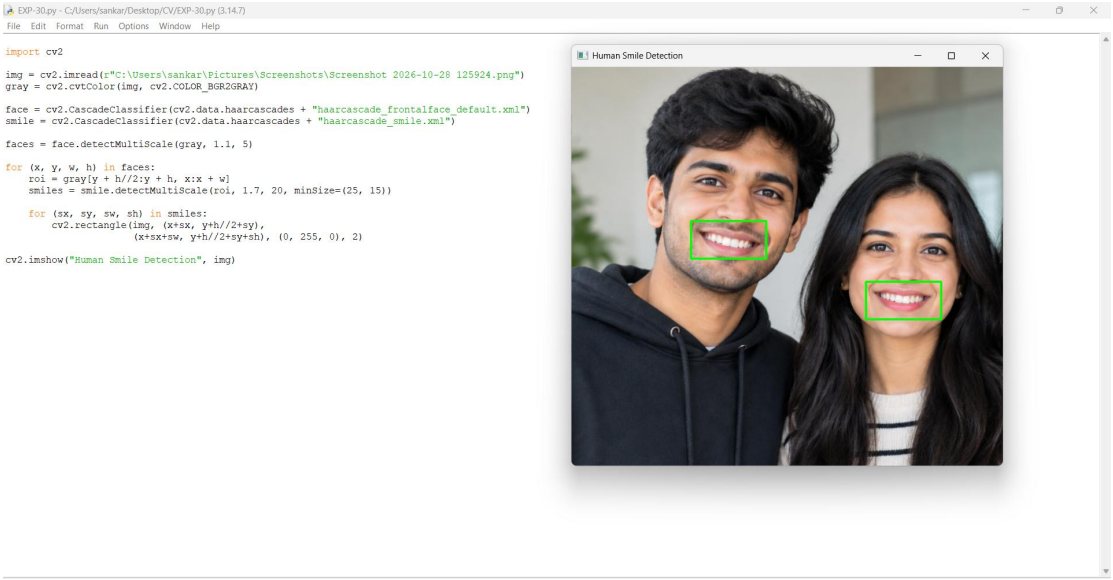
for (x, y, w, h) in faces:
    upper = gray[y:y + h//2, x:x + w]
    eyes = eye.detectMultiScale(upper, 1.1, 8, minSize=(20, 15))

    for (ex, ey, ew, eh) in eyes:
        cv2.rectangle(img, (x+ex, y+ey), (x+ex+ew, y+ey+eh), (0, 255, 0), 2)

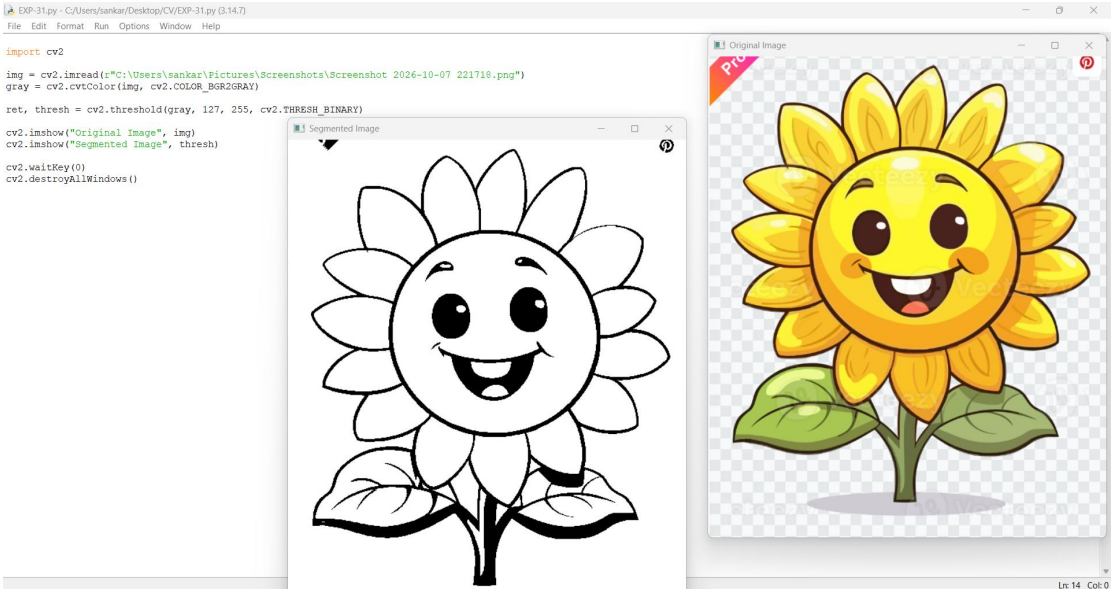
cv2.imshow("Human Eye Detection", img)
```



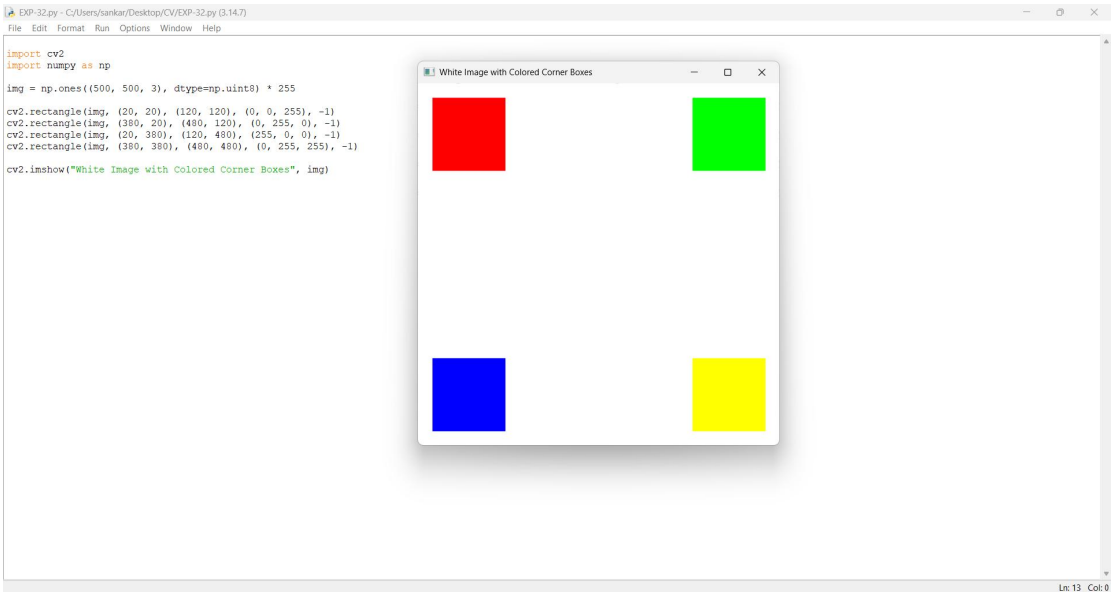
EXP-30: Implementation of Human Smile Detection using OpenCV



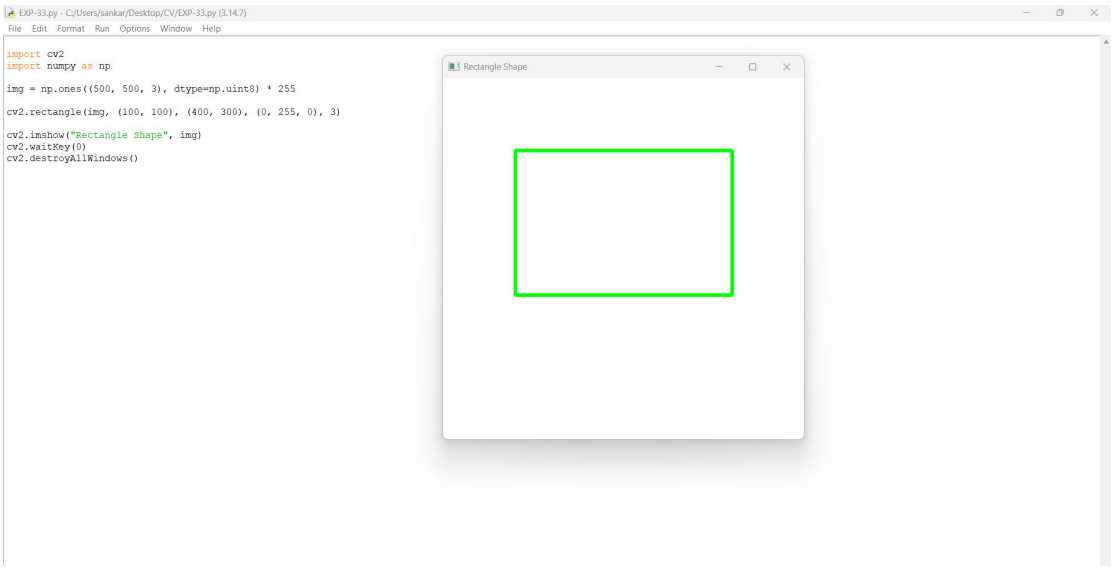
EXP-31:Implementation of Image Segmentation using Thresholding using OpenCV



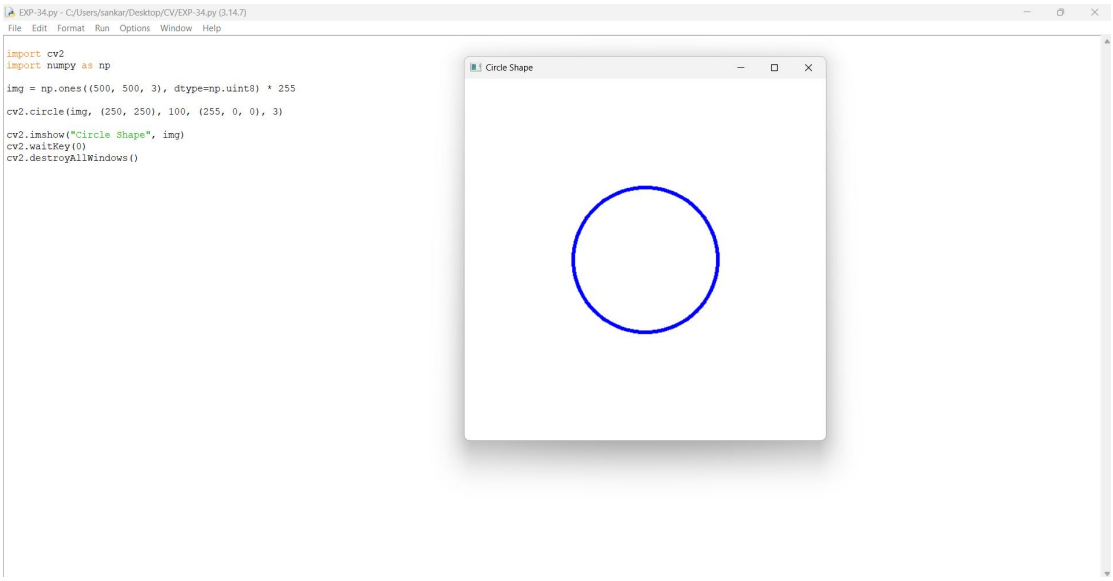
EXP-32: Creation of White Image with Four Colored Corner Boxes using OpenCV



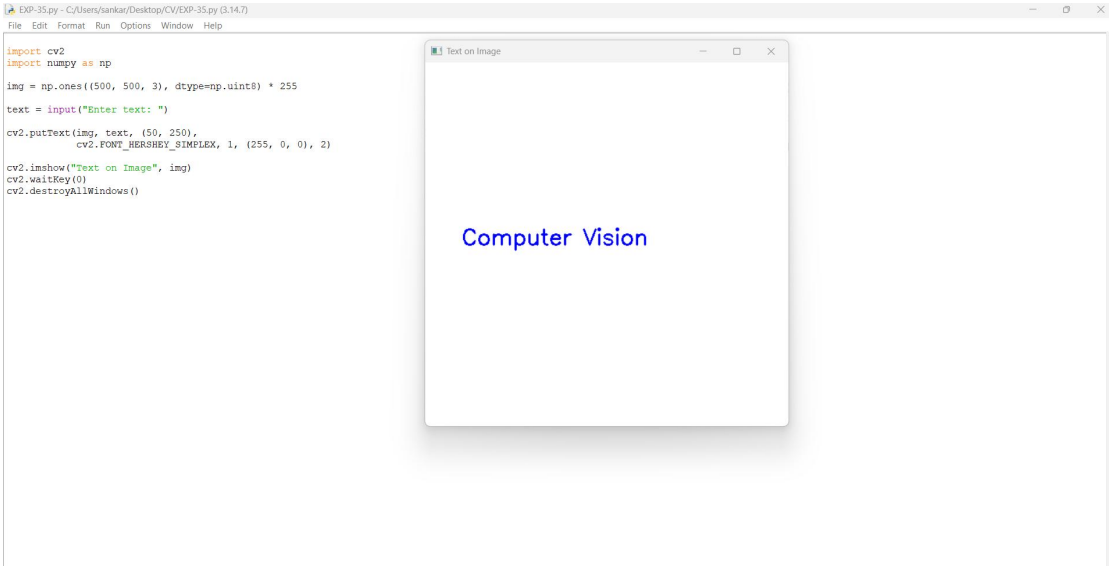
EXP-33 : Creation of Rectangle Shape using OpenCV



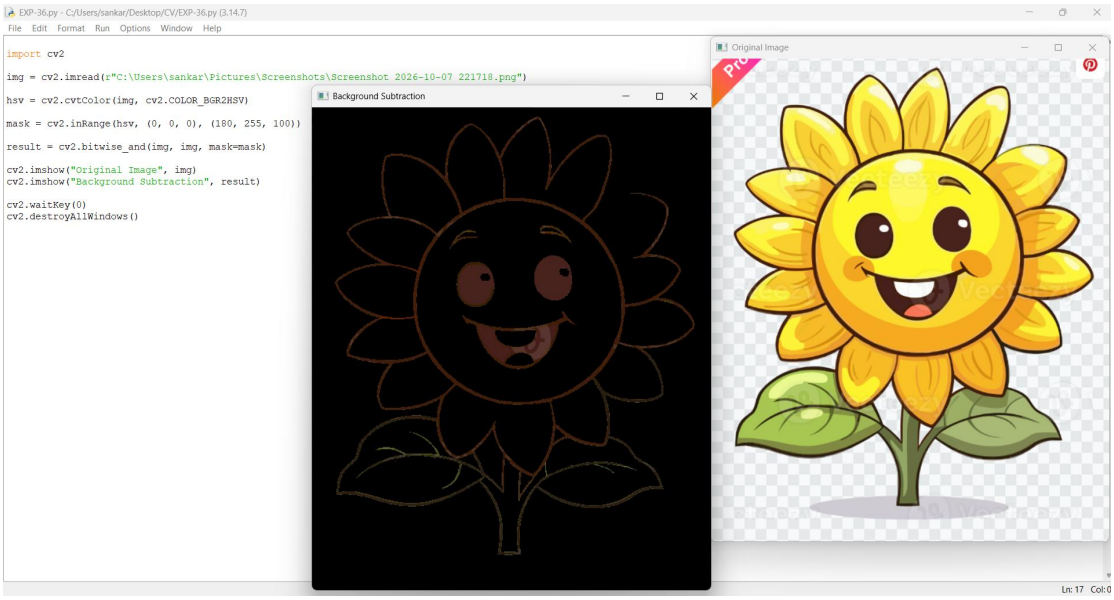
EXP-34 : Creation of Circle Shape using OpenCV



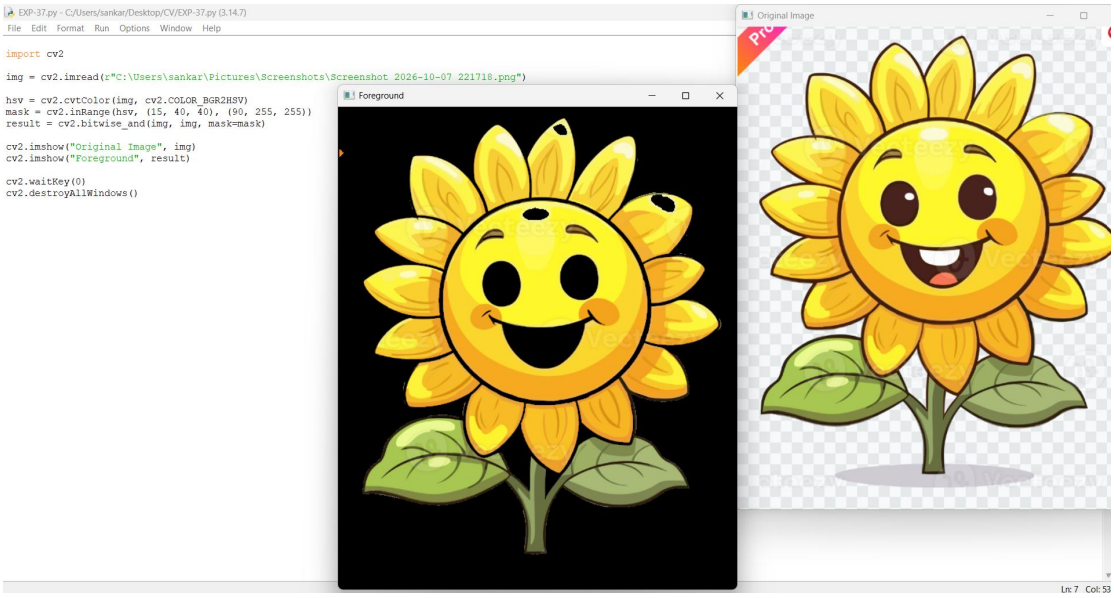
EXP-35: Display User Entered Text String on Image using OpenCV



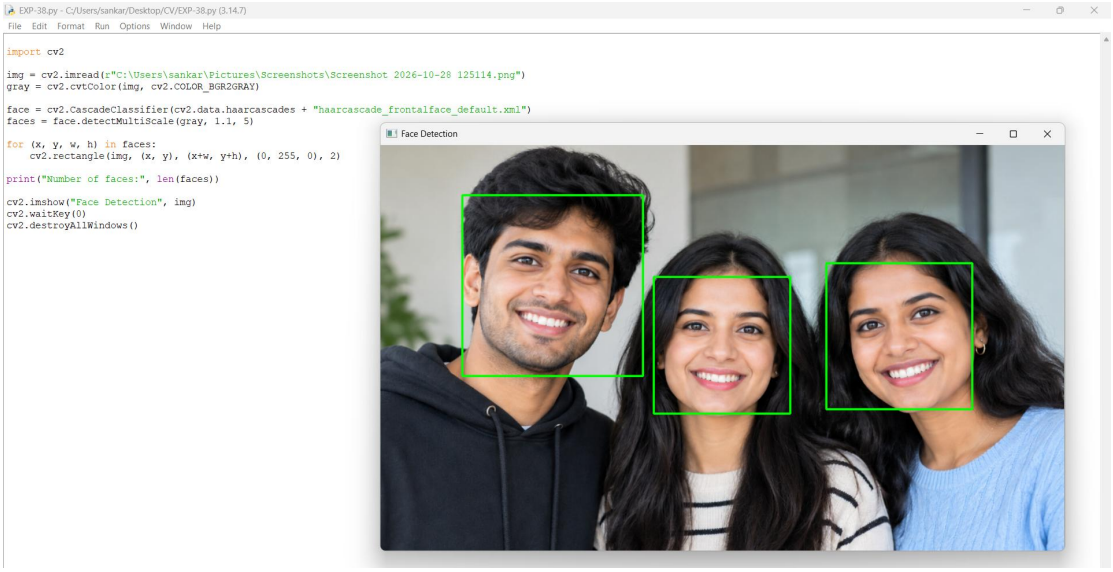
EXP-36:Background Subtraction based on Color Levels using OpenCV



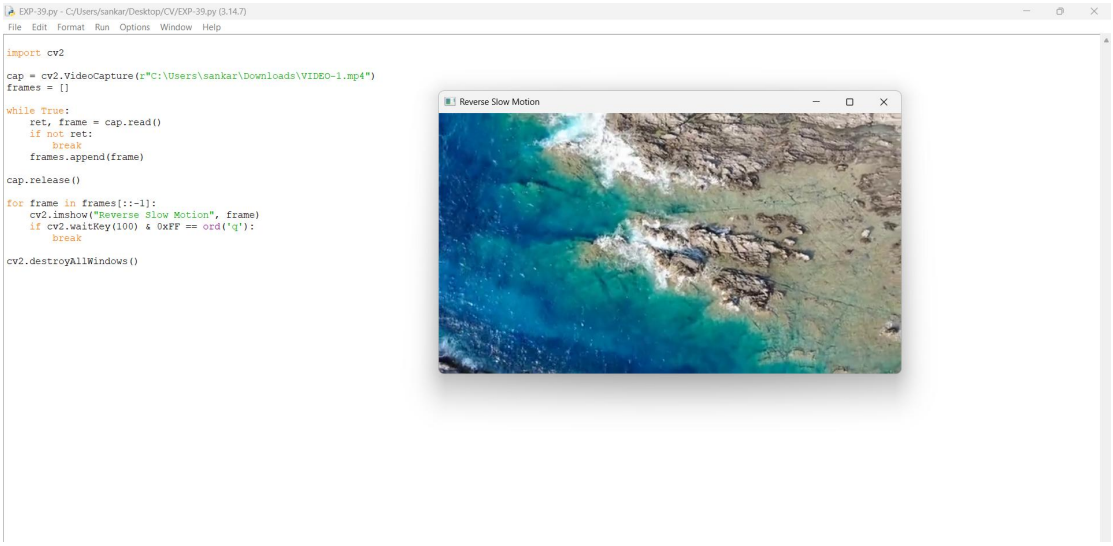
EXP-37: Foreground Subtraction based on Color Levels using OpenCV



EXP-38: Count the Number of Faces in an Image using OpenCV



EXP-39: Play Video in Reverse Mode with Slow Motion using OpenCV



EXP-40 : Extract Text from Videos using OpenCV

